Sze Chai Leung (Mickey)

Pasadena, CA 91106 | (217) 904-8882 | sleung@caltech.edu | szechaileung@outlook.com

EDUCATION

California Institute of Technology (Caltech) September 2022 - present
Ph.D. Candidate in Mechanical Engineering
M.S. in Mechanical Engineering (June 2024)

University of Illinois at Urbana-Champaign (UIUC) August 2018 - May 2022
B.S. in Mechanical Engineering with a Minor in Computer Science

- **GPA:** 3.98/4.00

PUBLICATIONS & PREPRINTS

1. **Leung, S. C.**, Zhou, D., & Bae, H. J. (2026). Reinforcement learning control of gust-induced airfoil lift fluctuations. *Manuscript in preparation*.
2. **Leung, S. C.**, Zhou, D., & Bae, H. J. (2026). Data-driven sensor placement for predictive applications: a correlation-assisted attribution framework (CAAF). *Under review at Communications AI & Computing*. https://arxiv.org/abs/2510.22517
3. Liu-Schiaffini, M., Singer, C. E., Kovachki, N., **Leung, S. C.**, Schneider, T., Bae, H. J., Azizzadenesheli, K., & Anandkumar, A. (2026). Tipping point forecasting in non-stationary dynamics on function spaces. *Under review at Proceedings of the National Academy of Sciences (PNAS)*. https://arxiv.org/abs/2308.08794
4. Huang, X., **Leung, S. C.**, & Bae, H. J. (2026). Consistency requirement of data-driven subgrid-scale modeling in large-eddy simulation. *Physical Review Fluids*, 11(1), 014602. https://doi.org/10.1103/yykb-6rvf
5. **Leung, S. C.**, Zhou, D., & Bae, H. J. (2024). Integrated gradients for optimal surface pressure sensor placement for lift prediction of an airfoil subject to gust. *AIAA Aviation Forum and ASCEND 2024* (p. 4148). https://doi.org/10.2514/6.2024-4148
6. Huang, X., **Leung, S. C.**, Whitmore, M. P., Elnahhas, A., & Bae, H. J. (2024). Consistent data-driven subgrid-scale model development for large-eddy simulation. *Proceedings of the Summer Program (pp. 395–404)*. Center for Turbulence Research, Stanford University. https://web.stanford.edu/group/ctr/ctrsp24/v04_HUANG.pdf
7. Ha, J., Kim, Y. S., Li, C., Hwang, J., **Leung, S. C.**, Siu, R., & Tawfick, S. (2023). Polymorphic display and texture integrated systems controlled by capillarity. *Science Advances*, *9*(26), eadh1321. https://doi.org/10.1126/sciadv.adh1321

RESEARCH EXPERIENCE

Bae Research Group (https://bae.caltech.edu/) Pasadena, CA
Advisor: H. Jane Bae, Assistant Professor of Aerospace September 2022 – present

Data-Driven Flow Control & Sensing for Gust Mitigation on Airfoils

- Apply deep reinforcement learning (RL) and conduct large-eddy simulations (LES) to design active control strategies that mitigate airfoil lift fluctuations
- Proposed a machine learning-based attribution framework to optimize sensor placement for target predictions across complex, nonlinear systems

Data-Driven Turbulence Modeling

- Developed subgrid-scale stress models via sparse regression coupled with a neural network to correct residuals and numerical errors, enhancing model consistency in LES

Ewoldt Research Group (https://ewoldt.mechanical.illinois.edu/) Urbana, IL
Advisor: Randy H. Ewoldt, Alexander Rankin Professor May 2021 – July 2022
Neural Network Constitutive Models for Non-Newtonian Fluids

- Formulated and fine-tuned neural networks to learn the physical properties of complex fluids from simulated flow fields

Kinetic Materials Research Group (https://tawfick.mechse.illinois.edu/) Urbana, IL
Advisor: Sameh Tawfick, Professor; Ralph A. Andersen Faculty Scholar May 2021 – October 2021
Capillary-Controlled Micro-Fin Morphing

- Tested and modeled the morphing behavior of polymer micro-fins under varying conditions

WORK EXPERIENCE

Mitsubishi Electric Research Laboratories Cambridge, MA
Research Intern June 2026 – present

- Integrate Computational Fluid Dynamics (CFD) solver with RL framework, establishing a training pipeline for active flow control optimization
- Apply and evaluate the RL framework across diverse flow control tasks

Department of Mechanical and Civil Engineering, Caltech Pasadena, CA
Graduate Teaching Assistant, Fluid Mechanics September 2023 – December 2023

- Delivered review lectures, conducted office hours, and managed assignments and exams

Mindray Medical International Limited Shenzhen, China
Mechanical Development Engineering Intern June 2020 - August 2020

- Led the design project of a single-person loading device for more convenient transportation of ultrasound machines
- Conducted mechanical testing on ultrasound prototypes and constructed engineering drawings

CONFERENCE PRESENTATIONS

1. **Leung, S. C.**, Zhou, D., & Bae, H. J. (2026). Optimizing sensor placement in turbulent flows: a correlation-aware attribution framework. *3rd ERCOFTAC "Machine Learning for Fluid Dynamics" workshop*, Amsterdam, Netherlands.
2. **Leung, S. C.**, Zhou, D., & Bae, H. J. (2025). Optimizing sensor placement in turbulent flows: a correlation-aware attribution framework. *The 78th Annual Meeting of the APS Division of Fluid Dynamics*, Houston, TX.
3. **Leung, S. C.**, Zhou, D., & Bae, H. J. (2024). Integrated gradients for optimal surface pressure sensor placement for lift prediction of an airfoil subject to gust. *AIAA Aviation Forum and ASCEND 2024*, Las Vegas, NV.
4. **Leung, S. C.**, Huang, X., & Bae, H. J. (2024). Consistent subgrid-scale model development for large-eddy Simulation with data-driven sparse identification. *The 77th Annual Meeting of the APS Division of Fluid Dynamics*, Salt Lake City, UT.
5. **Leung, S. C.**, Zhou, D., & Bae, H. J. (2023). Optimal surface pressure sensor placement for lift prediction of an airfoil subject to gust. *The 76th Annual Meeting of the APS Division of Fluid Dynamics*, Washington, DC.

SKILLS

- Python, Fortran, MATLAB, C/C++ programming
- High-fidelity numerical simulations, parallel computing on HPCs, mesh generation
- ProE/Creo, Solidworks and Fusion 360 CAD modeling